



TechnipFMC Case Competition: Team 9

4/6/2024

Meet the Team



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Agenda

1. Case Overview
2. Executive Summary
3. Spend analysis
4. Performance Metrics Analysis
5. In-house manufacturing
6. Take 5 Moment
7. Product Schedule
8. Cost analysis

Case overview

TechnipFMC has received a contract to provide 40 next-generation "Subsea 2.0" Christmas Trees. Subsea Trees are equipment attached to the wellhead that monitors and controls the flow of hydrocarbons. A sourcing strategy is required to obtain the requisitions needed to acquire the parts that go into the tree to complete this contract.

The following are key dates for deliverables:

- September 1st, 2025: 10 Subsea 2.0 Christmas Tree
 - Components must arrive by August 1st for final assembly
- December 1st, 2025: 30 Subsea 2.0 Christmas Tree
 - Components must arrive by November 1st



Executive Summary

Our procurement strategy for the “Subsea 2.0” project emphasized operational efficiency and ethical sourcing, utilizing the key performance indicator metrics to achieve sustainable business practices. By deciding to manufacture our composite valves in-house, we have ensured that we capitalize on our competitive advantages compared to the relative market. By choosing strategic partnerships with our suppliers, particularly Romania for tubing hangars and wing blocks, Poland for gate valves and metal seals, and Brazil for wing blocks, we heavily emphasized assessing our supplier's ethics, capabilities concerning production, and overall reliability. In doing so, we hope to deliver a well-rounded, feasible solution that aligns with TMFC’s business standards.

 Realizing possibilities We strive for ever better We take initiative We learn from success and failure	 Achieving together We work as one team We share knowledge We embrace diversity of thought	 Building trust We listen to improve We partner constructively We seek to outperform
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In everything we do, we never compromise on:

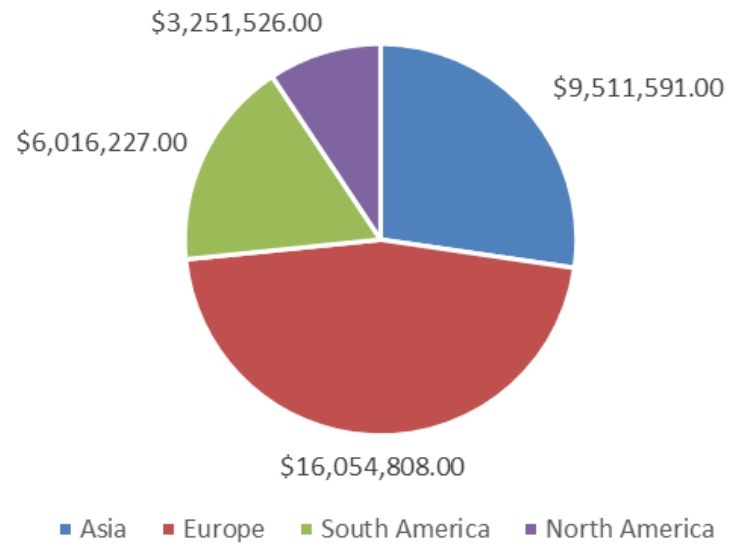
Safety	Integrity	Quality	Respect	Sustainability
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Spend Analysis

Historical Spend

- \$34,834,152 in the last 3 months

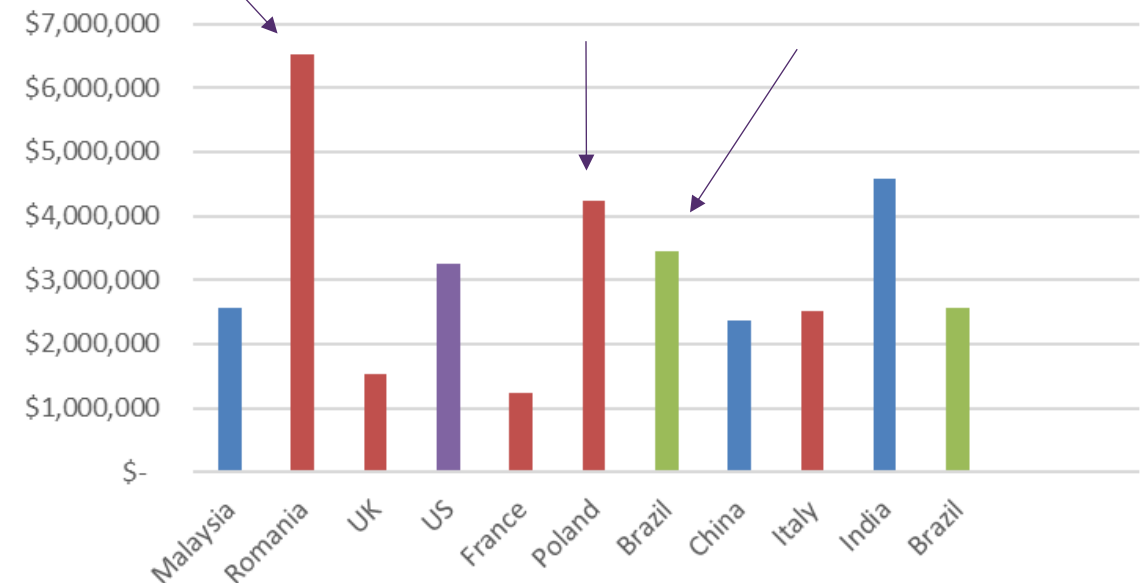
Cost Distribution By Region



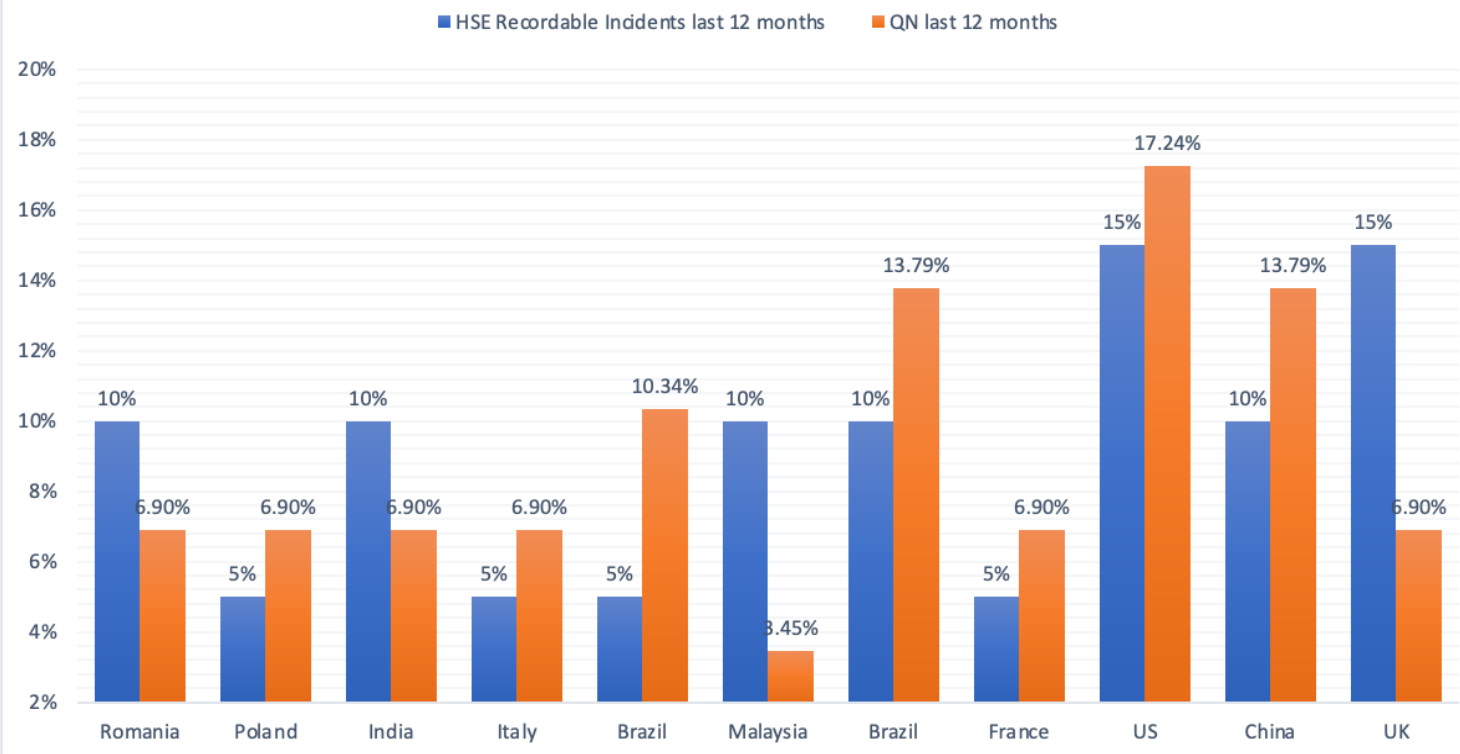
Category Spend:

- All parts: Machining

Spend Receipts Per Supplier



Overveiw of HSE and QN of Each Supplier



Safety and Quality: Top Priority

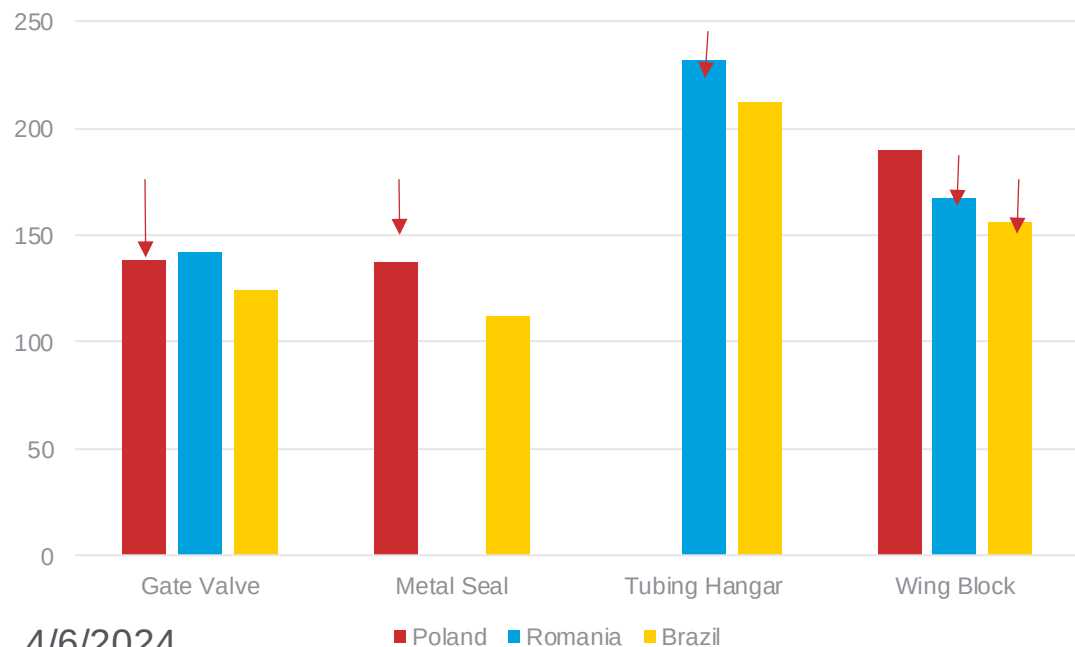
- Higher percentages suggest a greater number of incidents and there may be ongoing safety concerns or lapses in compliance.
- A few suppliers reported incidents, indicating that safety incidents are not isolated to a single supplier, ex. US
- We aim for a contractual agreement with suppliers containing a mutual goal of prioritizing complete safety, ensuring zero incidents throughout our collaboration.

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Max Lead times (days) for Aug 1st delivery-

Gate Valve: 153
 Metal Seal: 243
 Tubing Hanger: 243
 Wing Block: 166

Lead Time + Transit Time + Criticality (if needed)



Supplier Lead Times

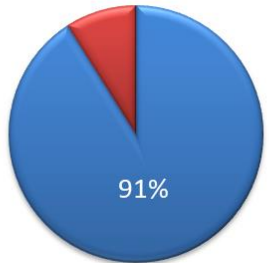
Time till August 1st		166 days
Suppliers	Country	Lead time
Supplier 1	Malaysia	216
Supplier 2	Romania	167
Supplier 3	UK	194
Supplier 4	US	180
Supplier 5	France	181
Supplier 6	Poland	190
Supplier 7	Brazil	156
Supplier 8	China	186
Supplier 9	Italy	181
Supplier 10	India	213
Supplier 11	Brazil	

*lead times
in days

- Brazil is the only supplier with a viable lead time of less than 166 days for the first batch of wing blocks
- Due to this, we will split the shipment between Brazil and Romania

On time delivery & SPI Rating

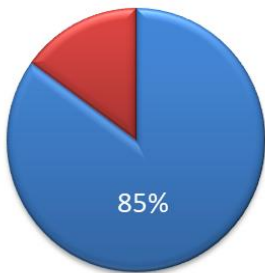
Supplier 6 (Poland)



Supplier 2 (Romania)



Supplier 7 (Brazil)



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Ensuring On Time Delivery

- OTD will be emphasized through scheduling and requirements in each suppliers Statement of Work to ensure correct timing
- Cushion time allocated for parts in the case of delivery delays

Supplier	SPI Score
Supplier 2	1.11
Supplier 6	1.12
Supplier 10	1.15
Supplier 9	1.17
Supplier 7	1.19
Supplier 1	1.2
Supplier 5	1.33
Supplier 11	1.33
Supplier 4	1.35
Supplier 8	1.38
Supplier 3	1.46

SPI Rating must not go over 1.25

In-house Manufacturing: Composite Valves

- All composite valves will be produced in house in TechnipFMC South American Plant
- New design competitive advantage can lead to growth in the market
- Cost to manufacture factoring direct materials and labor: \$88,404.00
- 40 Composite Valves total \$3,536,160
- Fast response times, better flexibility, enhanced quality control: cost advantages
- Shipment expected to arrive comfortably before assembly date



Take 5 Moment

In everything we do, we never compromise on:
Safety | Integrity | Quality | Respect | Sustainability



Safety Starts From Within: Promoting Mental Wellness For A Safe Workplace

Making Mental Health A Priority

- It's easy to become overwhelmed during a business project, especially when lots of data needs to be processed and analyzed.
- Poor mental health can cause employees to lose focus, rise tensions, and cause a project to fail
- Without proper attention, mental health will only decline

Improving Mental Health In the Workplace

1. Take a step back and find something peaceful to do
2. Express feelings and frustrations to those close to you
3. Pay close attention to the behavior of those around you
4. Exercise, eat nutritious food regularly, and take care of your body
5. Ask for help

Projected Schedule

Component	Criticality	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
Composite Valve Block: In house	10	Load sales orders	Able to Place PO Nov 1st			Arrives Feb 23							Assemble 10 Trees			Assemble 30 Trees		
Gate Valve: Supplier 6 (Poland)	7		Engineering Activities - Part on hold			Able to Place PO March 1st					Arrives Jul 28							
Metal Seal: Supplier 6 (Poland)	8		Able to Place PO Nov 1st						Arrives Apr 18									
Tubing Hanger: Supplier 2 (Romania)	10		Able to Place PO Nov 1st								Arrives Jul 21							
Wing Blocks: Supplier 7 (Brazil)	5		Engineering Activities - Part on hold			Able to Place PO Feb 15th					Arrives July 21							
Wing Blocks Batch 2: Supplier 2 (Romania)	5		Engineering Activities - Part on hold			Able to Place PO Feb 15th						Arrives Aug 1						
							supplier lead times, transit times, criticality times (if required)											

Cost Breakdown

Table 1

Supplier	Part	Quantity	Unit Price	Price
Poland	Gate Valve	80	\$32,500	\$2,600,000
Poland	Metal Seal	200	\$3,500	\$700,000
Romania	Tubing Hangar	40	\$87,050	\$3,482,000
Romania	Wing Block	60	\$8,600	\$516,000
Brazil	Wing Block	20	\$10,100	\$202,000

Table 2

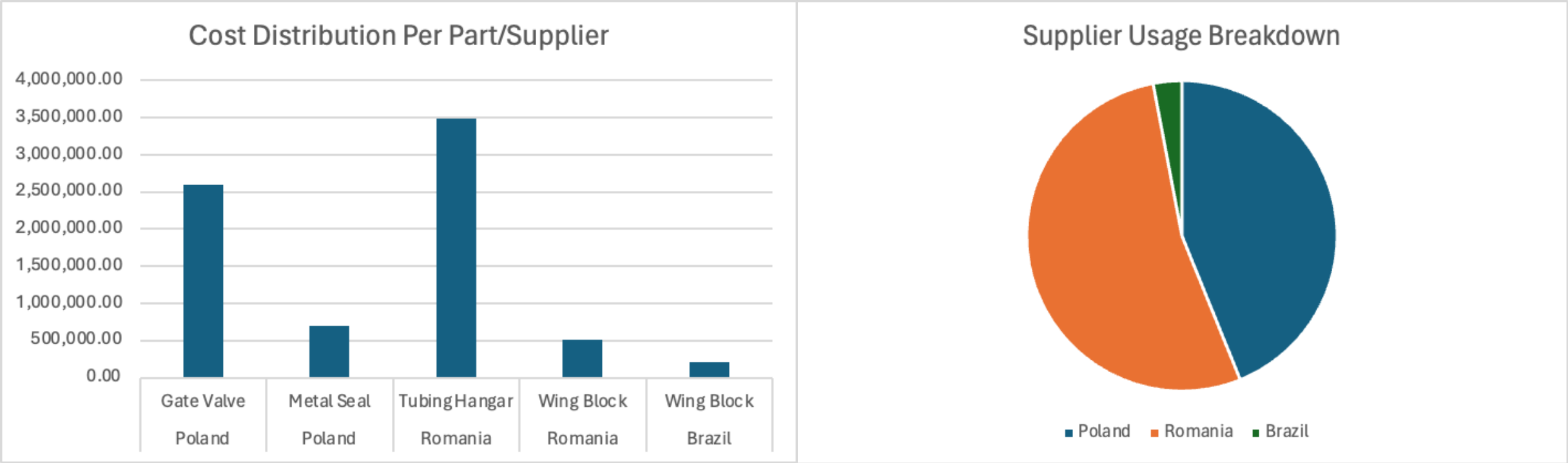
Supplier	Cost	Transit Cost	Discount	Total Cost
Poland	\$3,300,000	\$66,000	(\$330,000)	\$3,036,000
Romania	\$3,998,000	\$79,960	(\$399,800)	\$3,678,160
Brazil	\$202,000	\$4,040	\$0	\$206,040

Total Cost for sourced components: \$6,920,200

Total Cost for in-house components: \$3,536,160

Total Contract Cost: \$10,456,360

Cost Breakdown



Key Points:

- Wing Blocks are being sourced from two separate suppliers due to time constraints
- 44% of products are being sourced from Poland, 53% from Romania, and 3% from Brazil

Implementation and Follow Up plan

- **Contract Approval & Purchase Order Generation**

- Here, the procurement team negotiates with the selected suppliers and obtains any necessary approvals. After, they generate purchase orders ensuring that they align with contract agreements. The legal team also reviews to ensure compliance.

- **Receive materials**

- The logistics team coordinates with suppliers to determine transit times and costs to ensure timely deliveries. Supplier quality team also inspects upon receipt of materials to ensure they meet quality standards.

- **Pay & Performance Measure**

- Accounts payable is responsible the processing of supplier invoices. The supply chain analyst team tracks performance metrics related to delivery time, and lead times.

